

BCDR Final Project Scenario: Cyber Attack on SolveTech Inc. (Scenario Analysis)

SolveTech Inc., a medium-sized software development company specializing in cloud-based applications, has experienced a sophisticated ransomware attack. This attack has encrypted critical databases, including sensitive client information, proprietary software code, and operational data, leading to a complete system shutdown and rendering their IT infrastructure inaccessible. Based on the framework described in *Business Continuity & Disaster Recovery Planning for IT Professionals*, this event aligns with a **"Major disaster or disruption,"** as it "disrupts all or most of the normal business operations of the company and all or most of its critical business processes" due to "all or a majority of systems and equipment have failed or are inaccessible" [Snedaker, Chapter 7]. This elevates the incident to an extremely high threat level for SolveTech's operations, financial stability, and reputation, activating their full BC/DR plan.

Potential Impacts on Company Operations:

The ransomware attack will have an immediate impact on SolveTech's operations, significantly disrupting the function of their business.

- **Operational Paralysis:**
 - Software Development Halt: Encryption of proprietary software code and development environments will immediately cease all ongoing software development projects. This means developers cannot work on or deploy applications; this will directly impact core activities for a software development company.
 - Client Service Disruption: Access to sensitive client data will be compromised, preventing SolveTech from delivering services, providing support, or engaging in client communication. This could lead to breaches of Service Level Agreements and severe dissatisfaction, especially for a company specializing in cloud-based applications where continuous service is expected.
 - **Financial/Administrative Impairment:** Critical operational data, including financial records, billing systems, and potentially payroll, will be inaccessible. This will harm

revenue, payments processes, and employee compensation, threatening the company's cash flow.

- **Internal Communication Breakdown:** If internal communication platforms are server-dependent and encrypted, effective crisis response and employee coordination will be severely hampered during this chaotic period [Snedaker, Chapter 7, "Working Your Plan"].
- **Data Compromise and Loss:**
 - **Sensitive Data Exposure:** The encryption of sensitive client data raises immediate concerns about data breaches and potential regulatory non-compliance, requiring prompt notification and response under various data protection laws.
 - **Proprietary Code Loss:** Loss or irreversible encryption of proprietary software code represents a loss of core intellectual property for SolveTech. This significantly undermines the company's competitive advantage, future innovation, and long-term viability in the software development market.
 - **Potential for Irrecoverable Data:** The success of data recovery depends entirely on the effectiveness of SolveTech's backup and recovery strategies, which themselves may have been compromised or inadequately implemented. This highlights a critical vulnerability in risk mitigation [Snedaker, Chapter 7, "Implement risk mitigation strategies"].
- **Financial Repercussions:**
 - **Significant Revenue Loss:** The inability to deliver applications or services will directly result in a substantial loss of revenue during the prolonged downtime characteristic of a major disruption. This emphasizes the importance of understanding the "value" impacts of the decision, as highlighted in decision analysis, where financial outcomes are a key value measure.
 - **Ransom Payment Consideration:**
 - SolveTech will face the difficult decision of whether to pay the ransom, which can be costly and offers no guarantee of data recovery or the prevention of any future attacks. This is a complex decision, where a **deterministic model** could be built to project potential financial outcomes, considering factors like "Revenue," "Setup Costs," and "Variable Costs" associated with different recovery paths [decision_analysis_ch11_ppt-1.pptx, slide 7,17, 27].
 - **Legal and Regulatory Fines:** Data breaches involving sensitive client information could lead to legal liabilities, lawsuits, and fines under data protection regulations.